

Position-Dependent Curvature for Hyperbolic Embeddings of Medical Ontologies

A Research Proposal

Abstract

Modern medical large language models represent disease concepts in flat Euclidean embedding spaces, despite the established result that tree-structured data cannot be embedded in Euclidean space without distortion that grows logarithmically with the size of the tree. Recent work has applied constant-curvature hyperbolic embeddings to medical ontologies (Beam et al., 2018; Yang et al., 2025; Kim et al., 2026), demonstrating substantial improvements over Euclidean baselines. However, these methods impose a single global curvature on heterogeneous ontologies whose local structural properties vary substantially across regions. We have empirically verified this heterogeneity in ICD-10-CM: across the 255 sections of the ontology with at least three internal diagnosis nodes, mean branching factor varies from 2.33 to 10.00 — a 4.29-fold range — and subtree size per node varies tenfold across chapters. We propose to derive and evaluate a position-dependent curvature $\kappa(x)$ for hyperbolic embeddings of medical ontologies, with $\kappa(x)$ determined by a smooth combination of local branching factor and local subtree size computed at section-level granularity. We frame the embedding manifold as a Cartan-Hadamard manifold of variable negative curvature, generalizing the constant-curvature Poincaré ball model. We hypothesize this method will reduce embedding distortion in the heterogeneously-branched regions of medical ontologies, enable lower-dimensional embeddings suitable for resource-constrained clinical deployment, and improve downstream performance on rare-disease retrieval — a clinical task where the relevant ontology regions exhibit the highest local branching density. This proposal outlines the mathematical framework, baseline replication strategy, and downstream evaluation plan, and reports preliminary empirical analysis confirming the heterogeneity that motivates the approach. Experimental results from the proposed method are forthcoming.

1. Introduction and Motivation

Rare-disease misdiagnosis is a documented clinical crisis. The average rare-disease patient sees roughly seven specialists over five or more years before receiving a correct diagnosis, with cascading effects on treatment outcomes and survival. A significant component of this failure mode is informational: clinicians rely on lookup tools and AI-augmented decision support that struggle to navigate the implicit hierarchy of disease classifications. When a clinician searches for a candidate diagnosis, current systems poorly surface adjacent diagnoses — diseases that branch off one or two steps away in the medical taxonomy — leading to systematic misses for rare conditions whose symptom patterns overlap with more common diseases.

Modern medical large language models inherit and amplify this limitation. These models represent disease concepts as dense vectors in flat Euclidean embedding spaces. However, medical ontologies — including ICD-10-CM, SNOMED-CT, and Mondo

— are fundamentally hierarchical, and a classical theorem in metric embedding establishes that trees cannot be embedded in Euclidean space without distortion that grows logarithmically in the size of the tree (Linial et al., 1995). Two diseases that are siblings in the medical taxonomy must therefore land either too close, causing confusion, or too far apart, failing to capture relatedness, when embedded in flat space. This is a geometric property of the embedding space, not a problem fixable by additional training data.

Hyperbolic geometry, with its exponential volume growth, is the natural setting for tree-structured data. Sarkar (2011) showed that any tree admits a $(1+\epsilon)$ -distortion embedding into the Poincaré disk in linear time. Nickel and Kiela (2017) demonstrated that five-dimensional Poincaré embeddings substantially outperform two-hundred-dimensional Euclidean embeddings on hierarchy reconstruction. Beam et al. (2018) ported this technique to the ICD-9 hierarchy on a sixty-three-million-member insurance claims dataset, demonstrating that two-dimensional Poincaré embeddings perform comparably to one-hundred-dimensional Euclidean embeddings on the task of reconstructing chapter structure.

Recent work has continued to extend hyperbolic embedding methods to medical contexts. Yang et al. (2025) propose hyperbolic transformer-based methods for SNOMED-CT, demonstrating gains over sentence-transformer baselines on out-of-vocabulary clinical queries. Kim et al. (2026), in the GRAIL system, combine hyperbolic patient-trajectory representations with LLM-based reranking for clinical reasoning under hallucination constraints. Across this body of work, however, the curvature parameter is global — typically $K = -1$ — and treated as a hyperparameter rather than as a quantity derived from the structural properties of the data.

Medical ontologies are not structurally uniform, and the heterogeneity is large enough to matter geometrically. The volumetric requirement at a node of branching factor b is, to leading order, exponential in b , implying that the optimal local curvature should depend on local structure. Forcing all regions of a heterogeneous ontology into a single global curvature is geometrically suboptimal, and the resulting distortion concentrates precisely in the regions where the ontology is least uniform — often the same regions where rare diseases sit. This proposal develops the case that medical ontology embeddings should use a position-dependent curvature $\kappa(x)$ where the local sectional curvature varies smoothly across the embedding manifold to match the local structural properties of the underlying ontology.

2. Background

2.1 Hyperbolic Geometry and Tree Embeddings

Hyperbolic space is the unique simply connected complete Riemannian manifold of constant negative sectional curvature. In the Poincaré ball model, the manifold is the open unit ball in n -dimensional Euclidean space equipped with the conformally rescaled metric:

$$g_x = (4 / (1 - ||x||^2)^2) \cdot g^E$$

where g^E denotes the Euclidean metric. The induced distance between two points u, v in the ball is:

$$d(u, v) = \text{arcosh}(1 + 2||u - v||^2 / [(1 - ||u||^2)(1 - ||v||^2)])$$

Volume in n -dimensional hyperbolic space grows exponentially with radius — $V(r)$ scales as $\exp((n - 1)r)$ — in contrast to polynomial growth in Euclidean space. This volumetric exponential growth is what enables low-distortion tree embeddings: the number of nodes in a tree at depth r grows exponentially in r , matching the available volume in hyperbolic space. Sarkar's construction provides an explicit linear-time algorithm for embedding any weighted tree into the Poincaré disk with worst-case distortion at most $1 + \epsilon$ for appropriately scaled edges.

2.2 Hyperbolic Embedding Learning

Nickel and Kiela (2017) introduced a method for learning Poincaré embeddings via Riemannian stochastic gradient descent. Given a set D of observed parent-child or related pairs, they minimize a softmax-style negative-sampling loss:

$$L(\Theta) = \sum_{(u,v) \in D} \log [\exp(-d(u, v)) / \sum_{v' \in N(u)} \exp(-d(u, v'))]$$

The Riemannian gradient on the Poincaré ball reduces to a rescaling of the Euclidean gradient by the inverse metric tensor:

$$\theta_{t+1} \leftarrow \text{proj}(\theta_t - \eta_t \cdot ((1 - ||\theta_t||^2)^2 / 4) \cdot \nabla^E L)$$

with the projection operator ensuring embeddings remain inside the open unit ball. The $(1 - ||\theta||^2)^2 / 4$ multiplier encodes the curvature of the embedding space.

2.3 Hyperbolic Embeddings in Medicine

Beam et al. (2018) applied Poincaré embeddings to the ICD-9 hierarchy on insurance claims data, showing that two-dimensional Poincaré embeddings recover the ICD-9 chapter, sub-chapter, and major-code hierarchy with quality comparable to one-hundred-dimensional Euclidean embeddings. Yang et al. (2025) extended hyperbolic methods to SNOMED-CT with out-of-vocabulary retrieval; Kim et al. (2026) used hyperbolic patient-trajectory representations as a structured prior for LLM clinical reasoning. In phylogenetic embedding, where similar geometric setups arise, Macaulay et al. (2023) found empirically that improvement saturates around $K = -1$ across a wide range of curvatures, suggesting that for relatively uniform tree structures the choice of curvature is forgiving. Medical ontologies are not uniform — as the next section establishes empirically — which is the gap this proposal addresses.

3. Empirical Motivation: Heterogeneity in ICD-10-CM

We parsed the FY2025 ICD-10-CM Code Tables tabular XML release from CMS into a unified tree with one synthetic root, 22 chapters, 296 sections, and 46,498 diagnosis nodes for a total of 46,817 nodes and 46,816 edges. We note that the often-quoted figure of 70,000+ ICD-10 codes refers to billable codes including encounter-modifier variants (initial, subsequent, sequela); for the hierarchical embedding task the relevant count is the 46,500 distinct clinical concepts represented in the tabular hierarchy.

Across the 10,775 internal nodes (nodes with at least one child), branching factor has mean 4.34, median 3, and maximum 34. Aggregated to chapter level, mean branching factor varies only modestly: from 3.67 (Chapter 10, Respiratory) to 5.32 (Chapter 17, Congenital malformations), a 1.45-fold range. Within-chapter standard deviations lie between 1.3 and 2.8, indicating that each chapter is internally fairly uniform around its own mean. Maximum branching factors by chapter vary more substantially, from 10 to 21 (a 2.1-fold range), reflecting that some chapters contain individual outlier nodes with high local branching density.

Aggregated to section level — granularity finer than chapter — the heterogeneity is substantially larger. Across the 255 sections with at least three internal diagnosis nodes, mean branching factor ranges from 2.33 to 10.00, a 4.29-fold range with median of 4.25 and standard deviation of section means of 1.10. Chapter-level smoothing masks real fine-grained structural variation; section-level statistics expose it.

We computed a complementary measure: total descendant count (subtree size) for every internal node, summarized by chapter. Maximum subtree size by chapter ranges from 37 (Chapter 1, Infectious and parasitic diseases) to 361 (Chapter 19, Injury and poisoning) — approximately a 10-fold range. Mean subtree sizes per chapter range from 5.7 to 11.2. Some chapters contain densely-packed sub-branches with hundreds of descendants beneath a single node; others have shallower, sparser structure. This depth-density variation is complementary to branching-density variation and captures a distinct geometric property: how much hierarchical volume sits beneath any given embedding point.

Together, these analyses establish that ICD-10-CM exhibits real, multi-dimensional heterogeneity in its hierarchical structure: branching heterogeneity at section granularity ($\sim 4\times$ ratio) and subtree-density heterogeneity at chapter granularity ($\sim 10\times$ ratio). Both forms of heterogeneity correspond to local geometric requirements that a constant-curvature embedding cannot simultaneously satisfy across all regions. This is the empirical motivation for the proposed approach.

4. Problem Statement

Let $T = (V, E)$ be a medical ontology with rooted tree structure. For each internal node $v \in V$, define local structural statistics $b(v)$ (branching factor) and $s(v)$ (subtree size). We seek an embedding $f: V \rightarrow M$ into a Riemannian manifold (M, g) such that:

(a) the graph distance d_T on the tree is approximated by the Riemannian distance d_g on M with low multiplicative distortion;

(b) the manifold (M, g) is a Cartan-Hadamard manifold so that geodesics are unique and exponential maps are global diffeomorphisms;

(c) the sectional curvature $\kappa(x)$ at $f(v)$ varies smoothly across M as a function of the local structural statistics (b, s) at v ;

(d) the resulting Riemannian optimization remains computationally tractable on commodity hardware.

Existing methods correspond to the special case $\kappa(x) \equiv \kappa_0 < 0$ constant, implicitly assuming $b(v)$ and $s(v)$ are independent of v . Our empirical analysis shows this assumption fails in ICD-10-CM. The novel content of this proposal is the construction, derivation, and empirical evaluation of a non-trivial $\kappa(x)$ for medical ontologies.

5. Proposed Approach

5.1 Mathematical Framework

We propose a conformally Euclidean variable-curvature manifold. Consider M as an open subset of $\mathbb{R}^n$ equipped with the metric:

$$g_x = (1 / \lambda(x)^2) g^E$$

where $\lambda(x)$ is a smooth positive function generalizing the conformal factor $(1 - ||x||^2)/2$ of the standard Poincaré ball. The sectional curvature of this metric is determined by λ via a quasi-linear elliptic operator; the inverse problem of recovering λ from a desired κ is solvable analytically in special cases or numerically otherwise.

5.2 Parametric Form for $\kappa(x)$

We propose deriving $\kappa(x)$ from a smooth function $\rho(x)$ of local structural statistics:

$$\rho(x) = \alpha \cdot b_{local}(x) + \beta \cdot \log(s_{local}(x))$$

where $b_{local}(x)$ is a smoothed kernel estimate of local branching factor in the embedding region around x , $s_{local}(x)$ is a smoothed estimate of local subtree density, and α, β are learned scalars constrained to maintain $\kappa(x) \leq 0$ everywhere. The target curvature is:

$$\kappa(x) = \kappa_0 \cdot (1 + \gamma \cdot \rho(x))$$

with $\kappa_0 < 0$ a baseline and $\gamma \geq 0$ controlling variation amplitude. This parametric form is justified by the volumetric requirement: for a node of branching factor b at depth d , the number of nodes at depth d is b^d , and to fit these without collision the local volume must grow at least as fast. Matching this growth to the volume growth of constant-curvature hyperbolic space gives the leading-order target $|K| \propto (\log b)^2$, with subtree size providing a complementary correction for depth-density. Full derivation of the parametric form including verification that it preserves Cartan-Hadamard conditions for the relevant parameter ranges is a planned deliverable.

5.3 Embedding Optimization

The Riemannian gradient update generalizes from the constant-curvature case:

$$\theta_{t+1} \leftarrow R_{\theta_t}(-\eta_t \cdot g^{-1}(\theta_t) \nabla^E L)$$

where $g^{-1}(\theta_t)$ now varies with position and R denotes a retraction onto M . We propose using either the exponential map $\exp_{\theta}$ when computationally feasible (typically requiring numerical integration of the geodesic equation) or a projection-based retraction otherwise. The negative-sampling loss from Nickel and Kiela carries over directly, with distance $d(u, v)$ computed via geodesic-equation integration in the variable-curvature metric. We will build on the geoopt library (Bécigneul and Ganea,

2019) for the Riemannian optimization infrastructure.

5.4 Distortion Analysis

A central theoretical contribution will be a distortion bound for variable-curvature embeddings of trees with non-uniform structural statistics. The hypothesis: for an ontology with branching-statistic variation $\text{Var}(b)$ at section granularity and subtree-size variation $\text{Var}(s)$ at chapter granularity, the distortion of a Sarkar-style construction with position-dependent $\kappa(x)$ is bounded above by a quantity that decreases monotonically as both variations increase — that is, variable curvature helps more exactly when the ontology is more heterogeneous. Formalization of this bound is a planned deliverable.

6. Methodology and Experimental Plan

6.1 Phase 1: Data Pipeline (completed)

Acquired and parsed FY2025 ICD-10-CM tabular XML from CMS into a unified tree (46,817 nodes, 46,816 edges). Computed branching and subtree-size statistics at multiple granularities, establishing the empirical heterogeneity reported in Section 3. Generated diagnostic visualizations and persisted the parsed tree along with per-node feature statistics for use in subsequent phases.

6.2 Phase 2: Baseline Replication

Faithful replication of Beam et al.'s constant-curvature Poincaré embeddings on ICD-10-CM. Training via geoopt or the original Nickel-Kiela implementation. Evaluation metrics: mean rank, mean average precision, in-category versus out-of-category distance ratios. Multiple embedding dimensions evaluated (2, 5, 10, 50). This phase establishes the constant-curvature baseline against which all subsequent improvements are measured. Target: performance comparable to Beam et al.'s ICD-9 results, with 2D Poincaré approaching 100D Euclidean quality.

6.3 Phase 3: Variable-Curvature Implementation

Derive $\kappa(x)$ from the parametric form in Section 5.2, with local statistics computed at section-level granularity. Implement training in the variable-curvature manifold using geoopt with custom metric and retraction. Compare against Phase 2 baseline on identical evaluation protocol, controlling for embedding dimension, optimizer, and hyperparameters. Report per-chapter and per-section results to localize where variable curvature helps and where it does not. Numerical stability near the manifold boundary is addressed through gradient clipping and careful initialization.

6.4 Phase 4: Downstream Evaluation on Rare-Disease Retrieval

Use the Human Phenotype Ontology (HPO) and Mondo Disease Ontology to construct a symptom-to-disease retrieval benchmark targeting Orphanet rare diseases. Embed both HPO terms and Mondo disease entities in the variable-curvature space. Evaluate retrieval accuracy (top-k recall, mean reciprocal rank) compared to constant-curvature and Euclidean baselines. The intended demonstration: rare diseases whose phenotypic

presentation overlaps with more common diseases are poorly retrieved by Euclidean and (to a lesser extent) constant-curvature methods, but are retrievable when the embedding geometry respects the local structural density of the rare-disease subtree.

6.5 Datasets

Open and instantly available: ICD-10-CM (CMS, public domain), Mondo Disease Ontology (CC-BY 4.0), Human Phenotype Ontology and HPOA annotations, Orphanet rare-disease ontology, MedQA, and Med-HALT. Free with registration: UMLS Metathesaurus. Credentialed access required: MIMIC-IV (PhysioNet), used as optional supplementary validation. SNOMED-CT is optional and used only as a stretch-goal alternative ontology.

6.6 Compute Requirements

All training is feasible on a single GPU. Phase 2 baseline replication requires only modest compute (T4-class GPU sufficient for embedding dimension up to 100). Phase 3 variable-curvature training is more expensive due to numerical geodesic integration; an A100 GPU on Google Colab Pro+ is sufficient for embedding dimensions up to 50, well within the relevant range for ontology embedding tasks.

7. Expected Contributions

If successful, this work will produce: (C1) the first empirical characterization of structural heterogeneity in ICD-10-CM at section-level granularity, with implications for what kinds of geometric methods are appropriate for medical ontology embedding; (C2) an explicit derivation of a position-dependent curvature function $\kappa(\mathbf{x})$ for hierarchical embeddings of medical ontologies, motivated by local branching factor and subtree-size statistics; (C3) an implemented and benchmarked variable-curvature Riemannian optimization framework for ontology embedding, building on `geoopt`; (C4) empirical demonstration of improved embedding quality on ICD-10-CM and Mondo over constant-curvature baselines, with the improvement concentrated in heterogeneously-branched regions where rare diseases tend to reside; (C5) open-source code, trained embeddings, and parsed ontology data released under permissive license.

8. Limitations and Risks

The principal mathematical risk is that the position-dependent metric defined by $\lambda(\mathbf{x})$ may not satisfy the Cartan-Hadamard conditions globally for all parameter values, requiring constraints on (α, β, γ) to ensure non-positive curvature everywhere. The principal computational risk is numerical instability in geodesic integration near regions of rapidly varying curvature, addressed through adaptive integration step size and gradient clipping. The principal empirical risk is that improvements over constant-curvature baselines may not generalize across all chapters or all embedding dimensions; this is addressed by reporting per-chapter and per-section results and by structuring experiments to isolate where variable curvature does and does not help. Throughout, the constant-curvature baseline is maintained as the fallback comparison, ensuring that even negative results contribute information about the structure of medical ontologies.

9. Timeline

Phase 1 (data pipeline): completed. Phase 2 (baseline replication): months 1–2. Phase 3 (variable-curvature derivation and implementation): months 3–5. Phase 4 (rare-disease retrieval): months 5–7. Writing and revision: months 7–8. Submission to Regeneron Science Talent Search and follow-on academic venues: month 9.

10. Conclusion

Hyperbolic embeddings have established themselves as the appropriate geometry for hierarchical medical data, with strong empirical results across ICD, SNOMED, and patient-trajectory representations. However, existing methods uniformly assume constant curvature, ignoring the substantial heterogeneity of real medical ontologies — heterogeneity we have empirically verified in ICD-10-CM, where section-level branching varies by a factor of four and chapter-level subtree density varies by a factor of ten. We propose to derive, implement, and empirically validate a position-dependent curvature method that respects this heterogeneity, and to demonstrate its utility for the high-value clinical problem of rare-disease retrieval. The empirical heterogeneity established in this proposal grounds the project in observed structural properties of the data rather than assumed ones.

References

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